

Ten Effective Linux Commands for Systems Administrators



Systems administrators need a bag of tricks to ensure that everything runs smoothly without any hitches. Linux has a fine set of utilities and commands to assist sysadmins in their task. Mastering these tools will take the efficiency levels of Linux admins to a whole new level.

GNU/Linux is one of the most popular operating systems for servers. Today, most of the operating systems use advanced and modern graphical user interfaces (GUIs) but the CLI (command line interface) is still popular. Using the CLI/scripts, you can automate complicated tasks and execute them in a repetitive manner. In this article, we will discuss the most common CLI utilities. If you are familiar with GNU/Linux and want to become more productive, then this article is for you.

1) *find*

Searching for a file in a file system is a very common task and we have to do it quite often every day. GNU/Linux provides the *find* command which searches for files in a directory hierarchy. Given below is the syntax of the *find* command:

```
find [STARTING-POINT] [EXPRESSION]
```

In the above example:

- **STARTING-POINT** represents the directory's location, from where the search will start. Note that if **STARTING-POINT** is omitted, then the search will begin from the current directory.
- **EXPRESSION** is evaluated to search the file. **EXPRESSION** can be a name, type, size, permission, owner, and so on.

Let us search for a file with a given name. Here, we'll be

using the file name as an **EXPRESSION**.

```
$ find src-dir -name hello.txt
```

In the above example, *src-dir* is the **STARTING-POINT** and *hello.txt* is the **EXPRESSION**. Here the *-name* option indicates that we are searching for the file by name. If you want to perform a case-insensitive search operation, then use the *-iname* option instead.

For the *find* command, **EXPRESSION** can be a pattern as well. Many times we want to search certain types of files like *.txt*, *.jpg*, *.mp3* and so on. The example below shows how to use a pattern in **EXPRESSION**:

```
$ find src-dir -name "*.txt"
```

The above example will search and list all *.txt* files recursively from *src-dir*.

We can use the file type as an **EXPRESSION** with the *find* command. For instance, use the command below to search only directories:

```
$ find src-dir -type d
```

In the above example:

- The *type* option indicates we are performing a search